VARĖNA ARTILLERY RANGE, LITHUANIA

Field Fortifications – “Redoubts”

The analysis shows a training fragment of a fortress front in the form of a ravelin (redan)



Saulius Zaura
LiDAR analysis, 3D models, visualisations · 2026

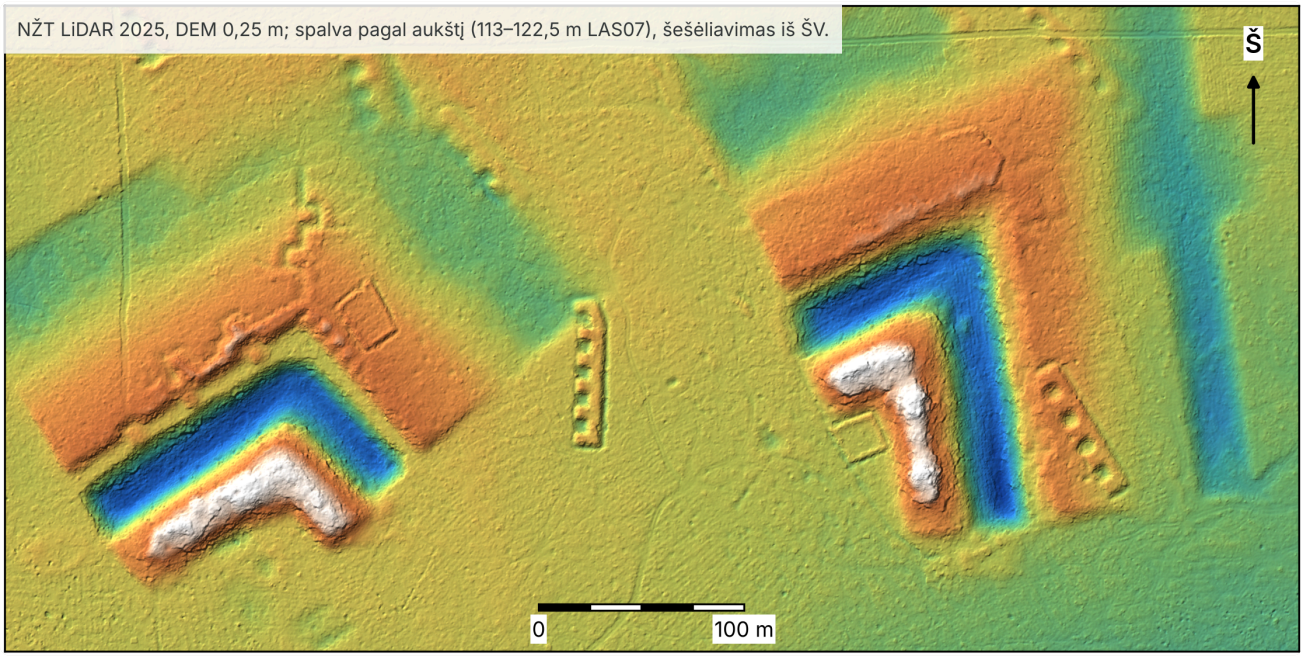
Created as part of the project: “Fort-Union – Integration of the Polish–Lithuanian Network of Fortifications for the Development of Representative Tourism Products”

Earthworks in the forest known as redoubts

In the pine forest of the former Varėna artillery range, near Salovartė, two large earthworks with deep ditches survive. Local tradition calls them “redoubts” (*redutai*). Airborne LiDAR data from 2025 made it possible to measure them beneath the forest and to reconstruct their original appearance in a 3D model.

The Orany (now Varėna) artillery range was laid out in 1877. Here the Imperial Russian army trained in gunnery and in building fortifications. The range was later used by the German and Lithuanian armies, and from 1931 the area became forest.

2	~30 m	~3.8 m	+4.9 m
earthworks	ditch width	ditch depth	rampart height



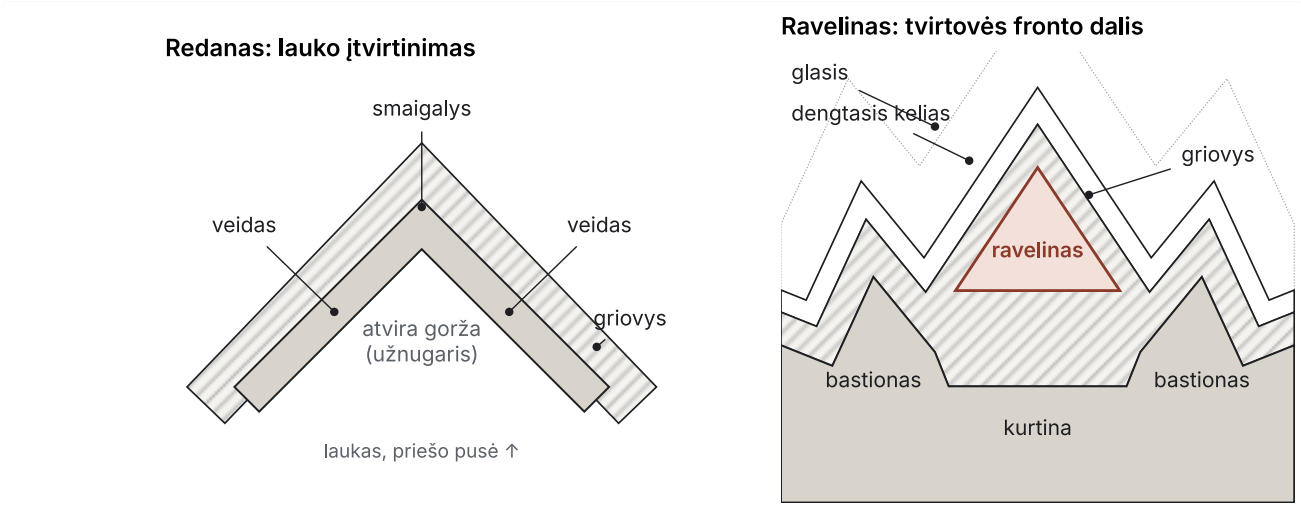
Present state from LiDAR 2025: blue shows the ditches, white the highest parts of the ramparts. (Map labels in Lithuanian: Š = north.)

> **Safety.** Unexploded ordnance may remain on the former range. Do not dig, and do not pick up metal objects.

Not a redoubt but a ravelin

A **redoubt** is a closed field work, defended on all sides. The Varėna earthworks are not closed: each consists of two faces meeting in a salient pointing north, with an open rear (gorge). This plan form is called a **redan**.

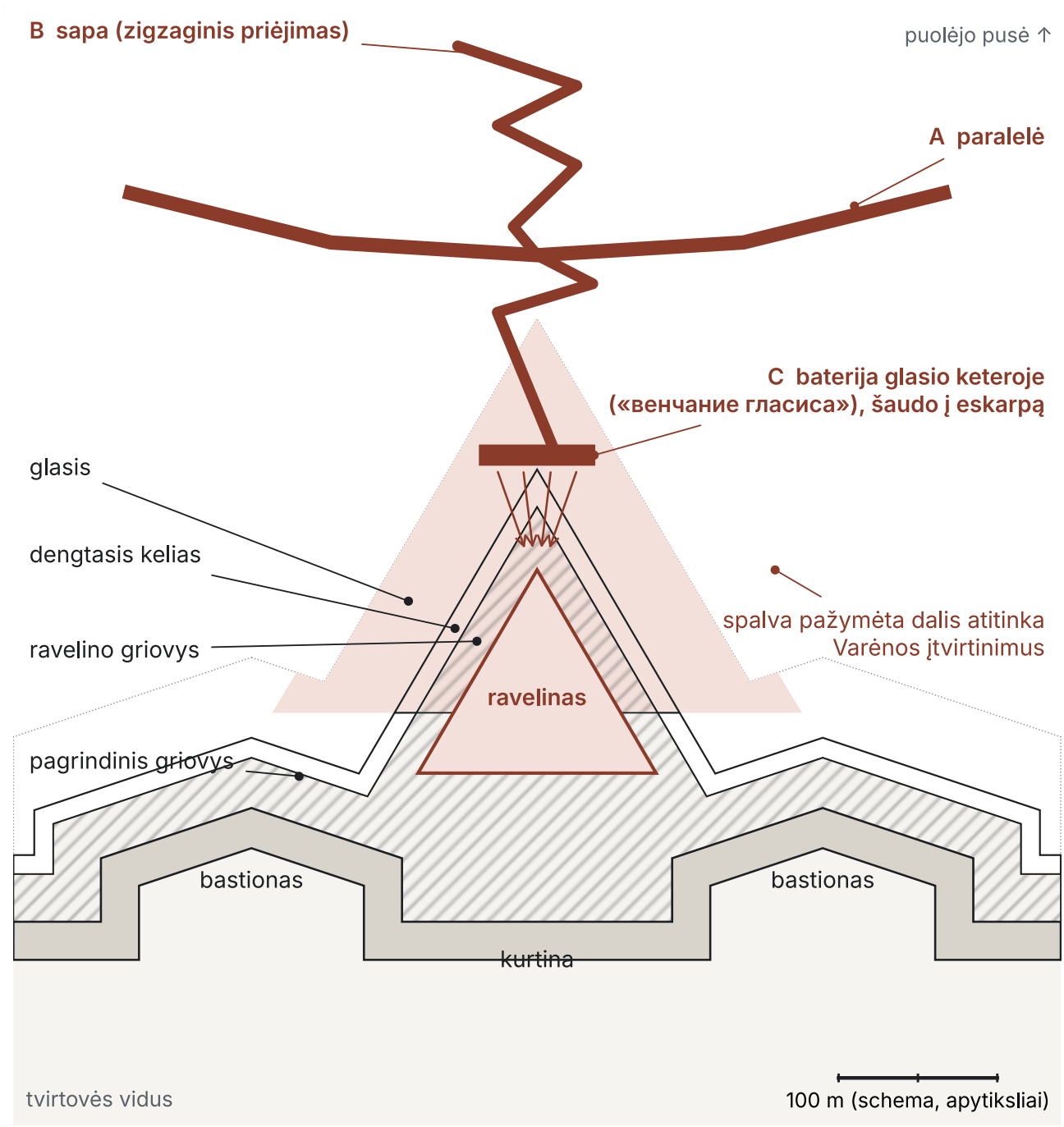
A field redan, however, is small: a rampart about 2 m high and a ditch about 2 m deep. At Varėna the rampart is twice as high, the ditch is about 30 m wide, and beyond it lie a **covered way** and a gently sloping **glacis**. These are elements of a permanent fortress. Among fortress works, this plan and profile belong to the **ravelin**, the outwork placed in front of the fortress rampart between two bastions.



Abu turi du veidus, smaigalį ir atvirą goržą. Ravelinas stovi priešais kurtiną ir ją dengia; prieš jį eina griovys, dengtasis kelias ir glasis. Varėnos įtvirtinimų pjūviuose išmatuoti būtent šie elementai (žr. profilius), todėl jie artimesni ravelinui nei lauko redanui.

Redan (left) and ravelin (right): the same plan form, but the ravelin has its own ditch, covered way and glacis. Diagram labels in Lithuanian (veidas = face, smaigalys = salient, griovys = ditch, dengtasis kelias = covered way, glasis = glacis, kurtina = curtain).

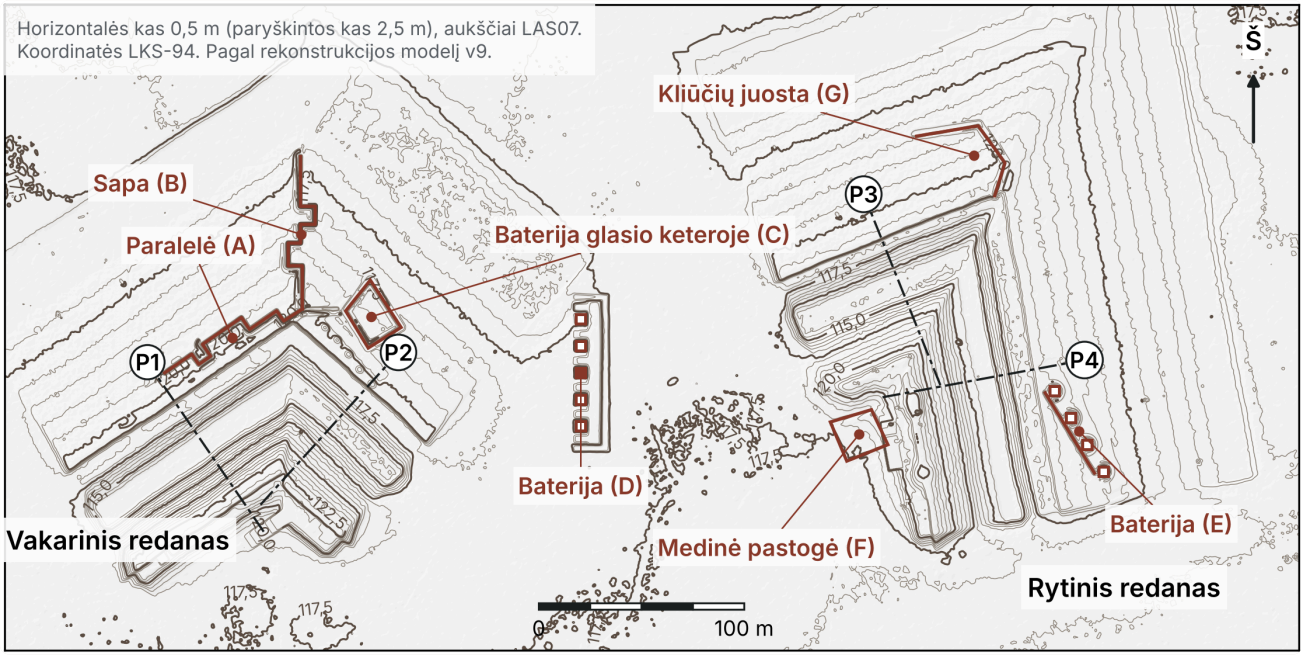
In front of the western earthwork, the besiegers' works survive: a zigzag approach trench (sap), a parallel trench and a battery on the crest of the glacis. This is the classic sequence of a fortress siege. We therefore conclude that this was a **training fragment of a fortress front**, where soldiers learned both to defend and to attack a fortress. No archival document describing these works has been found yet, so the conclusion rests on the measured form and on comparison with classical fortification.



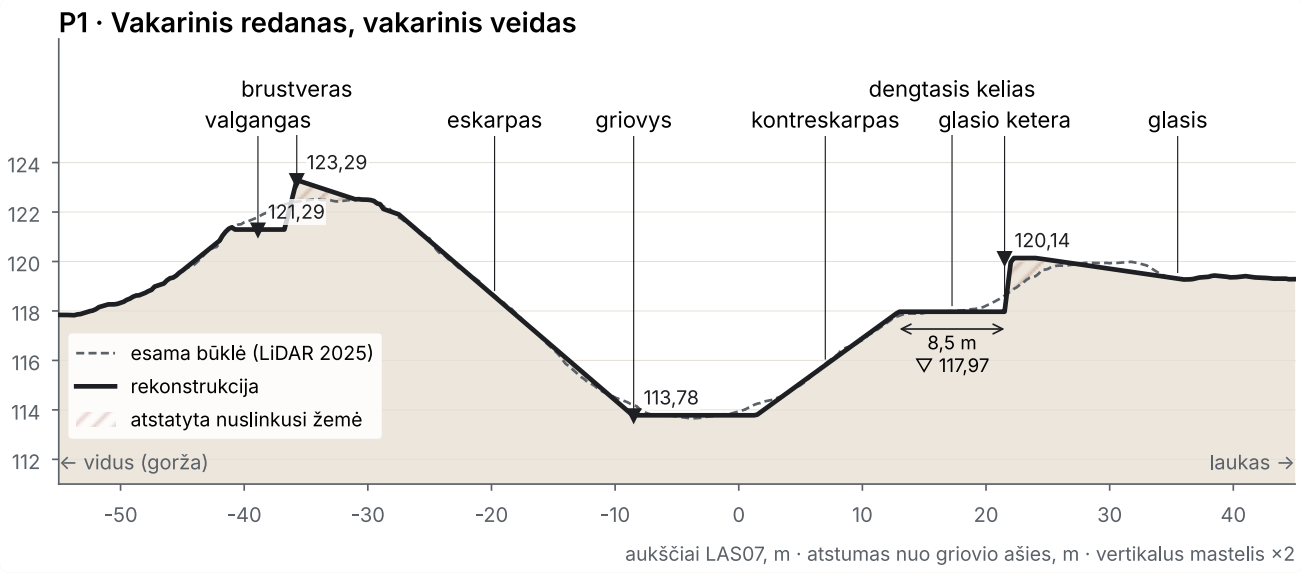
Bastioned fortress front and siege works. The coloured part corresponds to the Varėna earthworks: the ravelin with its ditch, covered way and glacis, and in front of it the parallel (A), the sap (B) and the battery on the glacis crest (C). Diagram.

How it looked during exercises

The reconstruction shows the works during Imperial Russian exercises: open ground without forest, maintained ramparts, defenders' and besiegers' batteries. What was measured, what follows analogues and what is a hypothesis is set out in the detailed report (in Lithuanian).

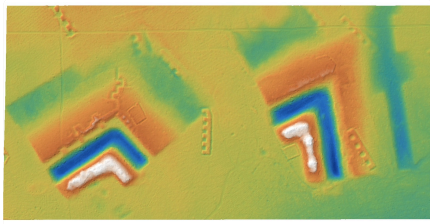


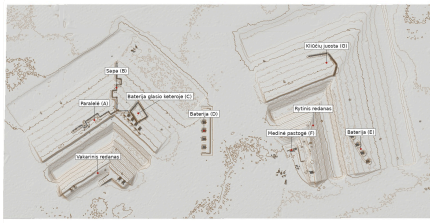
Reconstruction plan: western and eastern redans (vakarinis, rytinis redanas), parallel (A), sap (B), batteries (C, D, E), timber shelter (F) and obstacle belt (G).



Section through the western earthwork: present state (dashed) and reconstruction (bold). From left: terreplein (valgangas), parapet, escarp, ditch, counterscarp, covered way, glacis crest, glacis.

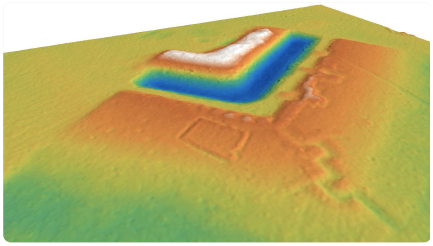


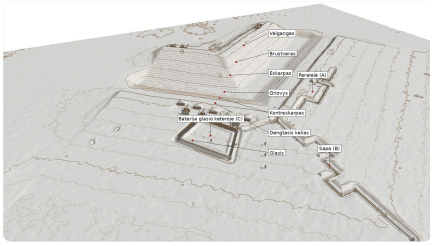




The whole site from above. Left: render refined with auxiliary AI tools; right: present state (LiDAR) and reconstruction drawing.







The western redan and the besiegers' works: battery on the glacis crest, sap and parallel.

Interactive 3D model

Reconstruction
skfb.ly/pOLGu

Scan the code to explore the earthworks in 3D on a phone or computer. The detailed report also includes a paper model: cut-outs to print, fold and glue.

© 2026 Saulius Zaura. Created as part of the project: "Fort-Union – Integration of the Polish–Lithuanian Network of Fortifications for the Development of Representative Tourism Products"
The shapes of the earthworks are reconstructed from measured LiDAR data. Images with the exercise scene are renders of the 3D model refined with auxiliary AI image-processing tools; people, guns and smoke in them are illustrative. LiDAR data: National Land Service of Lithuania, 2025. Textures and some objects: Poly Haven (CC0). "Cart and horse" model by depro777ru (CC BY 4.0). Historical material: L. Kasparaitė-Balaišė.

Interreg
Bendrai finansuoja EUROPOS SĄJUNGA
Lietuva – Lenkija